MILITARY SPECIFICATION

SWITCH ASSEMBLY: M78

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of switch assembly, designated as Switch Assembly; M78, which is used on the Sergeant warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDNANCE CORPS

8830291	- Switch Assembly: M78
EL-8854161	- Equipment List

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PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switch assembly shall comply with all requirements specified on the drawing (dwg) 8830291, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SMUPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Circuit resistance prior to activation. The resistance between the connector terminals prior to activation of the switch assembly shall be as specified in Table I when tested as specified in 4.4.2.

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Table I

From Terminal	To Terminal	Resistance (Ohms)	Test Code No.
J1 - t	J2 - AA	0.10 max.	
J1 - A	J2 - s	0.25 max.	030001
J1 - B	J1 - K	0.10 max.	030002
J1 - B	J1 - Z	0.10 max.	030003
J1 - B	J2 - P	0.10 max.	
J1 - C	J2 - Y	0.10 max.	
J1 - D	J1 - J	0.10 max.	
J1 - E	J1 - Y	0.10 max.	
J1 - F	J1 - H	0.10 max.	
J1 - G	J1 - X	0.10 max.	
J1 - L	J2 - f	0.20 max.	030004
J1 - M	J1 - N	0.10 max.	
J1 - M	J1 - P	0.10 max.	030005
J1 - R	J1 - q	0.10 max.	
J1 - S	J1 - p	0.10 max.	
J1 - T	J2 - E	0.10 max.	
J1 - T	J2 - J	0.10 max.	030006
J1 - T	J2 - L	0.10 max.	030007
J1 - U	J1 - f	0.10 max.	
J1 - V	J1 - n	0.10 max.	
J1 - W	J1 - e	0.10 max.	
J1 - a	J2 - i	0.10 max.	
J1 - b	J1 - k	0.10 max.	
J1 - c	J1 - g	0.10 max.	
J1 - c	J2 - N	0.25 max.	030013
J1 - d	J2 - e	0.10 max.	
J1 - h	J1 - i	0.10 max.	
J1 - j	J2 - t	0.10 max.	
J1 - m	J2 - W	0.10 max.	030008
J1 - r	J2 - y	0.10 max.	
J1 - s	J2 - BB	0.10 max.	
J2 - A	J2 - G	0.10 max.	
J2 - A	J2 - k	0.10 max.	
J2 - B	J2 - a	0.10 max.	
J2 - C	J2 - b	0.10 max.	
J2 - D	J2 - M	0.10 max.	
J2 - F	J2 - c	0.10 max.	
J2 - H	J2 - d	0.10 max.	
J2 - K	J2 - g	0.10 max.	
J2 - K	J2 - h	0.10 max.	
J2 - R	J2 - X	0.10 max.	
J2 - S	J2 - U	0.10 max.	
J2 - S	J2 - Z	0.10 max.	
J2 - T	J2 - j	0.10 max.	

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From Terminal	To Terminal	Resistance (Ohms)	Test Code No.
J2 - V	J2 - x	0.10 max.	
J2 - m	J2 - q	0.10 max.	030009
J2 - m	J2 - r	0.10 max.	030010
J2 - m	J2 - n	0.10 max.	030011
J2 - m	J2 - p	0.10 max.	030012
J2 - u	J2 - z	0.10 max.	
J1 - H	J1 - E	2.09 to 2.50	030014
J2 - j	J2 - i	5.60 to 6.35	030015
J1 - M	J1 - a	5.75 to 6.60	030016
J1 - T	J1 - g	5.75 to 6.60	030017
J1 - U	J1 - a	5.60 to 6.35	030018
J1 - m	J2 - N	5.60 to 6.35	030019
J2 - A	J2 - N	5.60 to 6.35	030020
J2 - D	J2 - N	5.60 to 6.35	030021
J2 - Y	J2 - X	5.60 to 6.35	030022
J2 - v	J2 - N	11.0 to 12.7	030023
J2 - w	J2 - N	11.0 to 12.7	030024

3.7 Insulation resistance prior to activation. The insulation resistance between each of the following terminals and case if metal encased and all other terminals listed below shall be 20 megohms minimum when tested as specified in 4.4.1.

J1 - A, C, D, E, G, L, R, S, V, W, a, b, d, g, h, j, r, s, t, and
J2 - B, C, F, H, K, S, V, m, and u.

Unused terminals J2 - CC, DD, EE, FF, GG, HH, JJ, KK, LL, MM, NN, and PP shall be connected together and tested as one terminal.

3.8 Circuit resistance after activation. The resistance between connector terminals after switch assembly has been activated in accordance with 3.11 shall be as specified in Table II when tested as specified in 4.4.2.

Table II

From Terminal	To Terminal	Resistance (Ohms)
J1-C	J1-G	0.10 max.
J1-D	J1-a	0.10 max.
J1-a	J1-c	0.10 max.
J1-R	J1-V	0.10 max.
J1-R	J1-b	0.10 max.

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From Terminal	To Terminal	Resistance (Ohms)
J1-S	J1-W	0.10 max.
J1-S	J1-h	0.10 max.
J1-T	J1-X	0.10 max.
J1-T	J2-K	0.10 max.
J1-T	J2-S	0.10 max.
J1-d	J1-p	0.10 max.
J1-j	J1-q	0.10 max.
J1-r	J2-u	0.10 max.
J1-s	J1-t	0.10 max.
J2-C	J2-D	0.10 max.
J2-B	J2-v	0.10 max.
J2-D	J2-H	0.10 max.
J2-F	J2-w	0.10 max.
J2-K	J2-W	0.10 max.
J2-T	J2-V	0.10 max.

3.9 Insulation resistance after activating. The insulation resistance between each of the following terminals and case if metal encased and all of the other terminals listed below shall be 20 megohms minimum when tested as specified in 4.4.1 after activation of the assembly in accordance with 3.11.
J1- A, K, L, M, P, b, h, m, r, and t.
J2- A, B, C, F, J, R, f, m, n, r, and s.

3.10 Diode characteristics. Forward voltage drop across the diodes and, in the reverse direction, across a series resistor of 100 ± 1 kilohm shall be specified. Measurements shall be made in accordance with 4.4.3.

Positive Lead to Terminal	Negative Lead to Terminal	Applied Forward Current	Max. Forward Voltage Drop	Code No.
J2-R	J2-P	400 ± 20 ma	1.0	037001

Positive Lead to Terminal	Negative Lead to Terminal	Applied Reverse Voltage	Max. Volt Drop Across Series Res.	Code No.
J2-P	J2-R	200 ± 10	0.3 (5)*	037002

*Figure in parenthesis is maximum voltage drop at 160°F.

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3.11 Activation. The squibs in the switch assembly shall be activated by the specified method when tested as specified in 4.4.4. The activation method, the associated terminal connector pins, and sequence of activation are specified in Table III. After activation the switch assembly shall comply with the requirements of 3.8 and 3.9.

Table III

From Terminal	To Terminal	Activating Method
J1-H	J1-E	4.4.4.1
J2-j	J2-i	4.4.4.2
J1-M	J1-a	4.4.4.2
J1-T	J1-g	4.4.4.2
J1-U	J1-a	4.4.4.2
J1-m	J2-N	4.4.4.2
J2-A	J2-N	4.4.4.2
J2-D	J2-N	4.4.4.2
J2-Y	J2-X	4.4.4.2
*J2-B	J2-N	4.4.4.2
*J2-F	J2-N	4.4.4.2

*These Squibs shall be activated after all other squibs.

3.12 Environmental testing. The switch assembly shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The switch assembly shall also comply with applicable paragraphs in section 3 before and after each environmental test. Testing shall be as specified in 4.4.5. Tests shall be conducted in the order specified.

3.12.1 High temperature. The switch assembly shall show no evidence of damage or deterioration cracks or swelling upon exposure to 160 ± 5°F and while at this temperature shall meet the requirements of 3.7 and 3.11 (omitting the requirement and test for 3.8) when tested as specified in 4.4.5.1. After the switch assembly has returned to ambient temperature, it shall comply with the requirements of 3.8, 3.9 and 3.10.

3.12.2 Low temperature. The switch assembly shall show no evidence of damage or deterioration such as distortion, cracks, or swelling upon exposure to -65 + 5°F and while at this temperature shall meet the requirements of 3.7 and 3.11 (omitting the requirement and test for 3.8) when tested as specified in 4.4.5.2. After the switch assembly has returned to ambient room temperature, it shall comply with the requirements of 3.8, 3.9 and 3.10.

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3.12.3 Transportation vibration. The switch assembly shall withstand vibration for 15 minutes in each axis shown in Figure 1 without evidence of damage affecting function when tested as specified in 4.4.5.3. The frequencies and amplitudes shall conform to Table IV. Following this test, the assembly shall meet the requirements of 3.6, 3.7 and 3.10.

Table IV

Freq. Range Hertz	Cycle Time	Constant Acceleration or Double Amplitude
8-55-8	7.5 Min	0.060 ± 0.006 Inch
55-500-55	7.5 Min	0.0 ± 1 Gravity (g)

3.12.4 Operational vibration. The switch assembly shall withstand without evidence of damage or affecting functioning a programmed sequence of band-limited white Gaussian noise and combined noise and sine wave vibration when tested as specified in 4.4.5.4. The total time of test shall be 210 seconds; 70 seconds in each of the vibration test will consist of the following five sections programmed continuously:

a. White Gaussian noise at a level of 22 g Root Mean Square (rms) band-limited between 15 and 1500 Hz for one (1) second.
b. A 4 g rms swept sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 800 to 1300 Hz at a rate increasing directly with frequency, one such sweep taking the full 15 seconds.
c. An 8 g rms swept sine wave superimposed on 15 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 15 seconds. The sine wave shall be swept from 1300 to 800 Hz at a rate decreasing directly with frequincy, one such sweep taking the full 15 seconds.
d. A 4 g rms sine wave superimposed on 7.5 g rms of white Gaussian noise band-limited between 15 and 1500 Hz for 38 seconds. The sine wave shall be swept from 15 to 1500 Hz and back to 15 Hz at a rate increasing directly with frequency. One such cycle shall take the full 38 seconds.
e. White Gaussian noise at a level of 22 g rms band-limited between 15 and 1500 rms for one (1) second.

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During section d of the vibration in the third axis, the switch assembly shall be activated as specified in 3.11. Following the vibration test, the assembly shall meet the requirements of 3.8, 3.9 and 3.10

3.12.5 Shock. The switch assembly shall withstand shocks of 30 ± 5 g three times in each direction along the three mutually perpendicular axes shown in Figure 1 for a total of 18 shocks when tested as specified in 4.4.5.5. Following the test, the assembly shall meet the requirements of 3.6, 3.7 and 3.10.

3.12.6 Humidity. The switch assembly shall withstand exposure to humidity of 95 ± 5 percent at 160 ± 5°F for three consecutive 24 hour test cycles without evidence of damage or affecting functioning when tested as specified in 4.4.5.6. Upon completion of the test the switch assembly shall meet the requirements of 3.6, 3.7 and 3.10.

3.13 Soldering. All soldering shall comply with the requirements of MIL-STD-1460.

3.14 Test codes (see 3.5). The contractor shall provide data feedback on punch cards in accordance with Quality Assurance Instruction QAI-PA-1. The data required to be coded, and the code numbers, are specified in the applicable paragraphs in this section.

3.15 Workmanship.

3.15.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.15.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.15.3 Burrs. Parts shall be free from burrs.

3.15.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.15.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.15.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8830291) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table V. Tests shall be conducted in order specified in Table V.

Table V

First article inspection and tests

Inspection or Test	Dest	Non-Dest	Unit 1	Unit 2	Unit 3
1. Classification of characteristics (see 4.3.2.1)		X	X	X	X
2. Circuit resistance prior to activation (see 3.6)		X	X	X	X
3. Insulation resistance prior to activation (see 3.7)		X	X	X	X
4. Humidity (see 3.12.6)	X		X		X
5. Shock (see 3.12.5)	X		X		
6. Transportation vibration (see 3.12.3)	X		X	X	X
7. Operational vibration (see 3.12.4)	X		X	X*	X*
8. High temperature (see 3.12.1)	X			X	
9. Low temperature (see 3.12.2)	X				X

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*Except 3.11, following operational vibration, the switch assembly shall meet the requirements of 3.6, 3.7 and 3.10 instead of 3.8, 3.9 and 3.10.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table VI and Table VII), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VII
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Switch Assembly, M78 prior to potting (dwg. 8830291).

Categories	Characteristics	Method of Inspection
Critical:		
	1. Incorrect soldering (see 3.14)	Visual
	2. Loose connection	Visual/Manual
	3. Damaged component	Visual
	4. Damaged wire insulation	Visual
Major:		
	101. Objective evidence relating to material missing or improper (see 3.1)	Visual
	102. Evidence of poor workmanship (see 3.12)	Visual
Minor:		
	201. Missing component	Visual
	202. Incorrect assembly (see 3.12)	Visual

4.3.2.1.2 Switch Assembly, M78 (dwg. 8830291).

Categories	Characteristics	Method of Inspection
Critical: None defined.		

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Categories		Characteristics	Method of Inspection
Major:			
	*101.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/32 inch in depth relative to the adjacent sockets)	Visual/Gage
	102.	Loose receptacles	Visual/Manual
	103.	Insufficient potting	Visual
	104.	Damaged components	Visual
	105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
	106.	Evidence of poor workmanship (see 3.12)	Visual
Minor:			
	201.	Inability of connector assembly to fit mating part	Gage
	202.	Location of mounting holes	Gage
	203.	Diameter of mounting holes, max	Gage
	204.	Missing component	Visual
	205.	Incorrect assembly (see 3.2)	Visual
	206.	Incorrect or illegible marking	Visual
	207.	Bare spot in protective coating except for slight scratches	Visual
	208.	Maximum external dimensions of assembly below brackets out of tolerance	Gage
	209.	Serialization improper or missing (see 3.4)	Visual

*In the event socket inserts do not appear to be within 1/32 inch in depth relative to the adjacent sockets, gage (see 4.3.4) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.032 inch maximum from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from measurements taken of sockets adjacent to the socket that appears to be defective. The forward metal edge of the connector shall be utilized at the reference surface for all measurements.

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4.3.2.1.3 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packaging and packing not in accordance with MIL-STD-1169 (MU)	Visual
102.	Marking not in accordance with MIL-STD-1169 (MU)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.12)	Visual

4.3.3 Testing. All tests shall be performed. Tests performed in the order specified below:

Test	Test Classification	Test Sequence	Group 1	Group 2
a. Circuit resistance prior to activation	Major	1	X	X
b. Insulation resistance prior to activation	Major	2	X	X
c. Diode characteristics	Major	3	X	X
d. Shock	Major	4		X
e. Transportation vibration	Major	5	X	X
f. Operational vibration	Major	6	X*	X
g. Low temperature	Major	7	X	

*Except 3.11, following operational vibration, the assembly shall meet the requirements of 3.6, 3.7 and 3.10 instead of 3.8, 3.9 and 3.10.
Of these tests, (d), (e), (f), and (g) are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit resistance prior to activation, insulation resistance prior to activation and diode characteristics. Each switch assembly shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with drawing requirements, the item shall be rejected and removed from the lot.

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4.3.3.2 Shock, transportation vibration, operational vibration and low temperature. Sample switch assemblies shall be selected at random in accordance with Table VIII. The sample switch assemblies shall be divided into two equal groups and each group of samples shall be subjected to these tests (see applicable paragraph in 4.4.5) as specified in 4.3.3. If any switch assembly fails to comply with the applicable requirement, the lot shall be rejected.

Table VIII

Sample sizes for destructive testing

Number Required to Be Delivered	Sample Size for No Failures
3-180	2
181-500	4
501-800	6
801-1300	8
1301-3200	10

Lot Size Required=Number Required to be Delivered + Sample Size.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8854161). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

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4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Insulation resistance. Insulation resistance shall be measured by means of an insulation resistance tester at 500 ± 50 direct current volts (VDC). The maximum current output of the insulation resistance tester shall not exceed 5 milliamperes (ma). For this test all the unused terminals of each connector shall be connected together and tested as one terminal.

4.4.2 Circuit resistance. The continuity and resistance shall be measured between the terminals listed in 3.6 and 3.8 with a four wire resistance measuring instrument. The maximum current output of the instrument shall not exceed 10 ma.

4.4.3 Diode characteristics. Diodes shall be tested for forward voltage drop when current specified in 3.10 is applied in the forward direction for not more than 15 seconds. Reverse current shall be determined by the voltage drop across a series resistor when the voltage specified in 3.10 is applied in the reverse direction for not more than 15 seconds.

4.4.4 Activation. Activate the squibs, using the methods of 4.4.4.1 and 4.4.4.2 as specified in 3.11. After activation of the squibs, the switch assembly shall be tested for compliance with the requirements of 3.8 and 3.9.

4.4.4.1 Activation by charged capacitor. Discharge a 70 ± 10 microfarad capacitor that has been charged to 35 ± 2 VDC by connecting it to the terminal connector pins specified in 3.11 to activate the squib in the circuit.

4.4.4.2 Activate by power supply. Connect to each set of terminal connector pins specified in 3.11, a power supply capable of supplying 3 amperes (A) minimum at 25 VDC maximum. Limit the current to 4 A maximum for activation of the squib in the circuit.

4.4.5 Environmental testing (see 3.11).

4.4.5.1 High temperature. The switch assembly shall be placed in a test chamber. The chamber's temperature shall be raised and stabilized at 160 + 5°F (see 6.4). After the switch assembly has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7 and 3.11 (omitting the test for 3.8). After the switch assembly has returned to ambient temperature, it shall be tested for compliance with the requirements of 3.8, 3.9 and 3.10.. Forced air shall not be used to aid in cooling the assembly.

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4.4.5.2 Low temperature. The switch assembly shall be placed in a test chamber. The chamber's temperature shall be lowered and stabilized at -65° + 5°F (see 6.4). After the switch assembly has been subjected to this stabilized temperature for a minimum of 4 hours and while still at this temperature, it shall be tested for compliance with the requirements of 3.7 and 3.11 (omitting the test for 3.8). After the switch assembly has returned to ambient temperature, it shall be tested for compliance with the requirements of 3.8, 3.9 and 3.10. Forced air shall not be used to aid in warming the assembly.

4.4.5.3 Transportation vibration. The switch assembly shall be mounted on a vibration machine and vibrated for 15 minutes in each of the three mutually perpendicular axes, as shown in Figure 1, at the frequencies and amplitudes specified in 3.12.3 with one 8 to 500 to 8 frequency cycle in 15 minutes. Total test time shall be 45 minutes. Upon completion of the vibrations, the switch assembly shall be tested for compliance with the requirements of 3.6, 3.7 and 3.10.

4.4.5.4 Operational vibration. The switch assembly shall be mounted on a vibration machine and vibrated in the three mutually perpendicular axes, as shown in Figure 1, using a standard noise or complex wave tape. The tape shall have the characteristics listed in the following table.

Time (sec)	Type of Signal	Ratio of Test Level To Calibration Level	RMS g Level
0-20	Noise (calibration)(1)	1.	Set gain to give 2 g rms
20-30	None	0	0
30-31	Noise (1)	11.	22.
31-46	Noise (1) plus sinusoid (2)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
46-61	Noise (1) plus sinusoid (3)	8.5	17.
	Noise alone	7.5	15.
	Sinusoid alone	4.	8.
61-99	Noise (1) plus sinusoid (4)	4.25	8.5
	Noise alone	3.75	7.5
	Sinusoid alone	2.	4.
99-100	Noise (1)	11.	22.

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(1) Band-limited (15-1500 Hz) white Gaussian Noise.
(2) Sine wave swept from 800 to 1300 Hz at a rate increasing directly with frequency. One full sweep in 15 seconds.
(3) Sine wave swept from 1300 to 800 Hz at a rate decreasing directly with frequency. One full sweep in 15 seconds.
(4) Sine wave swept from 15 to 1500 Hz at a rate increasing directly with frequency. One full cycle in 38 seconds.

For creating the random noise, the output of a General Radio random noise generator, or its equivalents may be used if proper care is taken to insure the correct amplitude distribution of the signal. For testing the noise generator or any other associated equipment, the use of an amplitude distribution analyzer is suggested. For insuring correct noise bandwidth, a filter with an asymptotic slope of at least 24 Decibel (Db) per octave and 3 Db points at 15 and 1500 Hz shall be considered acceptable. During section d of the vibration in the third axis (3.12.4), activate the switch assembly as specified in 3.11. Following the vibrations, test the switch assembly for compliance with 3.8, 3.9 and 3.10.

4.4.5.4.1 Method and procedure with the standard noise tape. Before applying the tape, the transfer function or frequency response of the entire shaking system shall be as flat as is consistent with the present state of the art. Plus or minus 3 Db or better over the band from 15 to 1500 Hz shall be considered as an objective in this respect. After proper equalization of the system, the standard noise tape is applied. The overall gain of the shake system is then adjusted so that with the noise calibration signal, the g level listed in 4.4.5.4 is obtained. With the same gain setting, the tape is run for the remainder of the test and the g levels obtained will conform to the levels listed in the table of 4.4.5.4.

4.4.5.5 Shock. The switch assembly shall be rigidly mounted to a test fixture as applicable which shall be mounted to a suitable shock producing apparatus. The switch assembly shall then be subjected to 3 shocks in each direction of the 3 major mutually perpendicular axes, as shown in Figure 1. Each shock shall be of 30 $\pm$ 5 g peak amplitude and 11 $\pm$ 2 milliseconds (ms) duration as measured between points which are 10 percent of the peaks. Time to peak shall be 5.5 $\pm$ 1 ms. After completion of the 18 shocks, the switch assembly shall be examined for damage and tested for compliance with the requirements of 3.6, 3.7 and 3.10.

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4.4.5.6 Humidity. The Switch Assembly, with its connectors covered with a potted mating connector, shall be placed within the test chamber and shall be subjected to the humidity test cycle as specified in MIL-E-5272, 4.4 Procedure I. All temperatures shall be maintained at ± 5°F. Three such cycles shall be performed consecutively. The pH value shall be adjusted by the use of chemically pure hydrochloric acid or chemically pure sodium hydroxide. Upon completion of the last cycle, the switch assembly shall be removed from the test chamber and, with the moisture removed by wiping, shall comply with the requirements specified in 3.12.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switch assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The switch assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The switch assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.4, 3.14 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

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6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2072 Rev. 1 dated 9 August 1965.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A158

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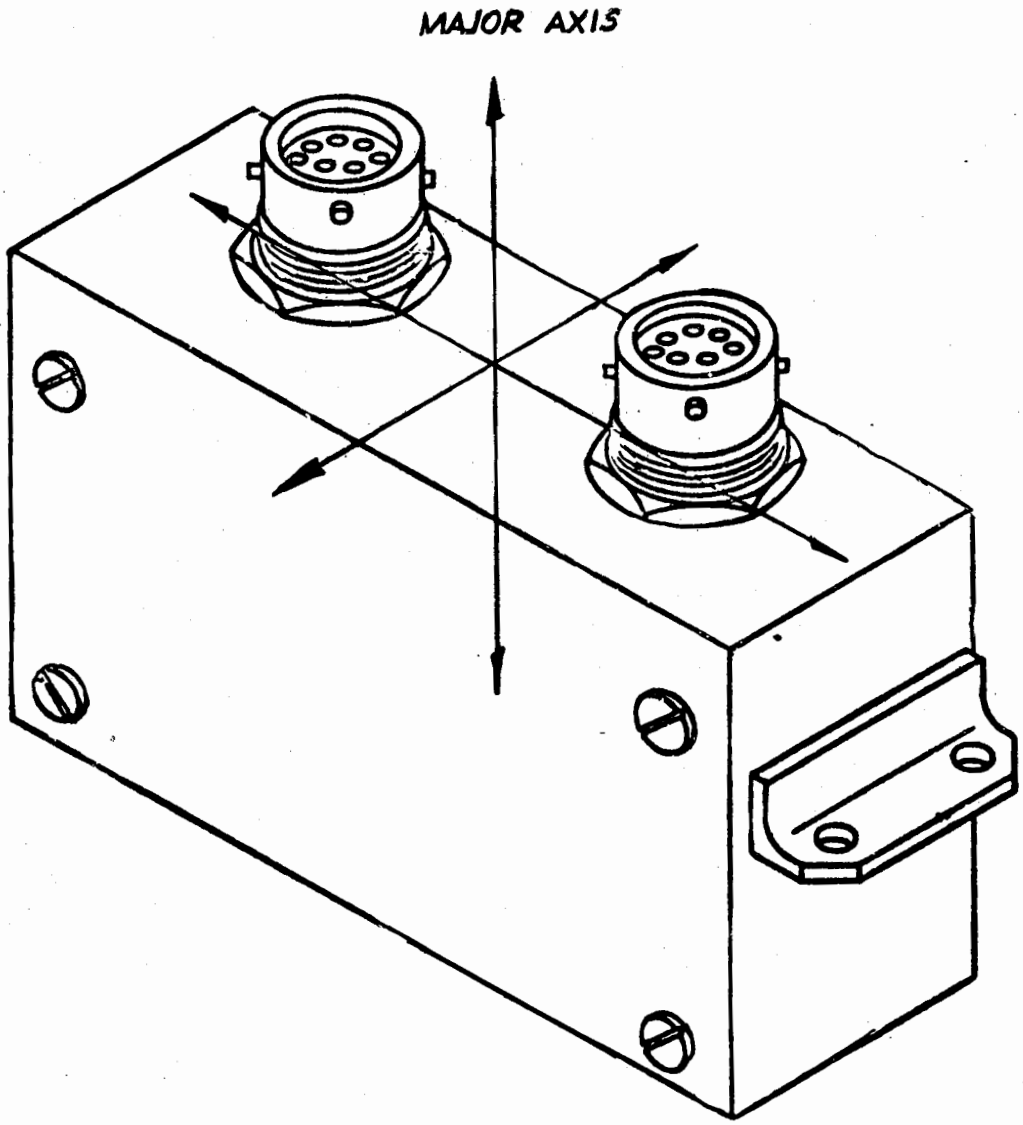


Figure 1 - Axes for vibration and shock

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